IBM SkillsBuild AI Experiential Learning Lab

Solution Design

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Optional

Intelligence & Reasoning Design

Purpose: Define the technical depth – the brains of the agent apart from just architecture and workflow.

When: By week 5

Knowledge & Retrieval

What the agent needs to know

At minimum, it should retrieve from these categories.

Docs

domain ontology or taxonomy / role definitions / skill dictionaries
tool dictionaries / task libraries / certification catalogs
evidence templates / starter-path templates / domain transition rules

Policies

minimum profile completeness before matching /confidence threshold for top-domain recommendation
when to ask clarification / when to offer a starter quiz
when to use template fallback / how many sources are required before recommending a role or certification
source precedence, for example internal verified data over scraped web data
explanation policy, such as “always show why this domain matched”

Historical data

previous user profiles and accepted domain choices
overrides, such as when users picked domain 2 instead of predicted domain 1
click-through on recommended roles / course/certification completion
placement or outcome data if available
feedback on whether suggested pathways felt relevant
missing-node history, so you can see where the graph repeatedly breaks

Primary RAG pattern:

Graph RAG + hybrid evidence retrieval + policy gating

Source stack:

graph DB for ontology
SQL/NoSQL for profiles and logs
vector + keyword index for evidence docs
APIs for certifications and labor-market freshness
config/rules store for thresholds and fallbacks

Retrieval rules:

retrieve graph neighborhood first
retrieve evidence second / rerank with domain metadata, recency, trust, and user goal alignment
confirm with user before locking context

Freshness rules:

official-source checks for dynamic content
stale-score penalties
scheduled refresh for external sources
manual review on ontology edits

Fallback rules:

starter quiz if profile incomplete
template fill if graph incomplete
explicit gap flagging instead of silent failure

Reasoning & Decisions

The agent makes a series of linked decisions: whether the user profile contains sufficient signal, which domains best match the user, which graph path should be traversed within the selected domain, whether the resulting subgraph is complete enough for downstream use, and whether fallback or clarification is required. Its reasoning combines classification, planning, evaluation, and multi-step logic. First, it classifies the user across all domains using weighted profile signals. Next, it plans by traversing the most likely domain subgraphs to identify roles, skills, tools, tasks, and evidence pathways. It then evaluates both the confidence of the match and the completeness of the graph structure. Confidence thresholds determine whether the system can recommend a domain directly, should present multiple candidates for user confirmation, or must request clarification. Validation logic checks input sufficiency, score separation, graph integrity, evidence support, and final path usability before the output is passed to downstream agents.

Multi-Agent Logic (if used)

A multi-agent architecture is adopted to decompose the complex task of career navigation into specialized, controllable components. Rather than relying on a single monolithic agent, the system separates profile interpretation, domain classification, graph traversal, validation, and evidence retrieval into distinct agents to improve accuracy, transparency, and robustness. The interaction pattern combines a sequential backbone, where each stage depends on the validated output of the previous step, with parallel execution for domain scoring, candidate path exploration, and evidence retrieval. A supervisory agent orchestrates the workflow, enforcing confidence thresholds, triggering clarification when profile signals are insufficient, and activating fallback strategies when graph data is incomplete. This hybrid design ensures both efficiency and reliability while preventing early-stage errors from cascading through downstream agents.

Risks & Constraints

The system faces several technical, data, and decision-making risks, including graph incompleteness, cascading errors from early-stage misclassification, retrieval inaccuracies, sparse user signals, and model hallucination. These risks are further amplified by the tradeoff between autonomous decision-making and supervised control. To mitigate these challenges, the system adopts a hybrid strategy combining graph validation, confidence-based decision gating, human-in-the-loop confirmation, and fallback templates for incomplete domains. A multi-agent architecture enables modular validation at each stage, while hybrid retrieval with freshness and source-priority controls ensures evidence quality. Confidence thresholds and margin-based ranking determine when the system operates autonomously versus when it solicits user input. This design balances efficiency with reliability, preventing error propagation while maintaining a usable and transparent user experience.

Governance & Ethics

Purpose: Explain how your AI agent is designed to behave *safely, responsibly, ethically* and in compliance with privacy and governance expectations. Include mechanisms that *prevent harm*, *protect users*, and *build trust*.

When: by Week 5

Ethical Considerations

The agent is designed as a safety-bounded, human-centered decision-support system rather than a fully autonomous decision-maker. It promotes fairness by restricting decision inputs to job-relevant signals and excluding protected characteristics from matching logic, while regular bias testing and counterfactual evaluation help identify and mitigate unfair patterns. It supports transparency through explainable recommendations, confidence indicators, rationale summaries, and auditable decision traces. User rights and dignity are protected by allowing clarification, confirmation, and override rather than imposing deterministic outcomes. Privacy and security are addressed through data minimization, purpose limitation, role-based access control, secure storage, retention controls, and privacy-aware logging. Accountability remains with the deploying organization and designated human overseers, who can review decisions, intervene when confidence is low, and investigate any failures or harms. Together, these measures help prevent harm, protect users, and build trust in the system.

Governance & Compliance

The AI agent operates under a structured governance and compliance framework that aligns internal approval processes with external trustworthy AI principles. Deployment and updates require multi-level approval from product, technical, and governance stakeholders, with formal change management and version control. The design incorporates fairness, accountability, and transparency by restricting decision inputs to relevant signals, providing explainable outputs, and maintaining audit logs for traceability. It aligns with established frameworks such as the OECD AI Principles and UNESCO's Recommendation on the Ethics of AI, emphasizing human-centered design, oversight, and data governance. Ongoing compliance is ensured through continuous monitoring of system performance, user overrides, and bias indicators, supported by periodic audits, logging, and incident response mechanisms. This governance structure ensures that the system remains reliable, ethical, and compliant throughout its lifecycle.

Scope & Limitations of the MVP

Purpose: Explain what your MVP will and will not do.
When: by Week 5

Tip: At the end, you only have 10 weeks. You will need to prioritise – focus on the 2-3 functionalities that will deliver the greatest user value for your MVP.

MVP Scope

The MVP focuses on delivering a reliable end-to-end experience that transforms a user's unstructured profile into a ranked set of domain matches and a clear, actionable starting path. By Week 10, the system will include profile interpretation, top-3 domain classification with explanations and confidence levels, a simplified domain graph with entry roles and key skills, and a user confirmation step to prevent cascade errors. Lightweight validation and fallback mechanisms ensure robustness when inputs are incomplete or data is missing. More advanced capabilities, such as full domain coverage, complex modeling, and large-scale data integration, are intentionally deferred to prioritize core user value: reducing uncertainty, providing direction, and building trust through transparent and explainable recommendations.

Limitations

The MVP is intentionally constrained to prioritize core functionality over completeness. It supports only a subset of domains with a partially populated graph, uses rule-based domain classification rather than advanced machine learning, and relies on a small curated dataset for evidence rather than large-scale or real-time data integration. Personalization, multi-agent orchestration, governance automation, and scalability are implemented in simplified forms or deferred. While these constraints may limit coverage, precision, and system sophistication, they enable rapid validation of the core value proposition—accurate, explainable domain matching and actionable career guidance—within a controlled and debuggable environment.

What is next

The solution will be improved through an iterative, staged development process from MVP to full-scale deployment. Initial testing focuses on validating core functionality, domain matching accuracy, and user alignment through synthetic cases and early user feedback. In subsequent phases, the system incorporates expanded domain coverage, improved models, feedback-driven refinement, and A/B testing to enhance precision and usability. At scale, the system integrates advanced retrieval, personalization, and multi-agent orchestration, supported by continuous monitoring, fairness auditing, and governance controls. This staged approach ensures that improvements are data-driven, validated, and aligned with user needs while maintaining reliability and ethical compliance.

✅ Checkpoint – Week 5

Tip: Validate technical decisions early with prototypes and stress tests. Ensure ethical and governance considerations are not an afterthought—they should be integrated into design from the start.

At this stage, you should be clear about the MVP that you are building and will submit at the end of the this program.

- ❑ Is the architecture modular, scalable, and aligned with solution goals? Does it supports performance, scalability, and maintainability?
- ❑ Are workflows clear, efficient, and user-centric? Does it aligns with user experience and business objectives?
- ❑ Are tools and integrations secure, compitble (interoperable), and cost-effective?
- ❑ Is the intelligence logic well-though and justified?
- ❑ Does the design ensures data privacy, fairness, explainability, and bias mitigation?
- ❑ Does the governance framework addresses compliance, and accountability?

Open Questions & Assumptions

What we don't know yet

List gaps in knowledge that need validation.

The most critical uncertainties are:

- **Domain matching accuracy in real-world settings**
- **Minimum input required for reliable recommendations**
- **Completeness and correctness of the domain graph**
- **User trust and explanation effectiveness**
- **Bias and fairness risks**
- **Practical usefulness of recommended paths**

Assumptions we are making

Document any assumptions about users, processes, or technology.

The most important assumptions underlying the MVP are:

- Users can provide enough input for meaningful matching
- Domain selection is the key decision point
- A simplified graph can represent career pathways adequately
- Top-3 domain recommendations provide sufficient guidance
- Users will engage with explanations and confirmation
- Rule-based logic is sufficient for early-stage accuracy

Validation research

List what tests are needed to gather information and/or validate assumptions

These tests collectively validate:

- ❑ **Accuracy** → domain matching, path quality
- ❑ **User behavior** → acceptance, trust, engagement
- ❑ **System robustness** → fallback, edge cases
- ❑ **Fairness** → bias detection
- ❑ **Design assumptions** → input sufficiency, explanation value
- ❑ **Architecture choices** → multi-agent vs simple pipeline

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Continue the journey

- ***Weeks 6-8:***
    - *Implement the technical work*
    - *Ensure depth & robustness*
    - *Carry out user testing & iteration*

- ***Weeks 9-10:*** *Prepare and finalise your project submission*

Good Luck!!